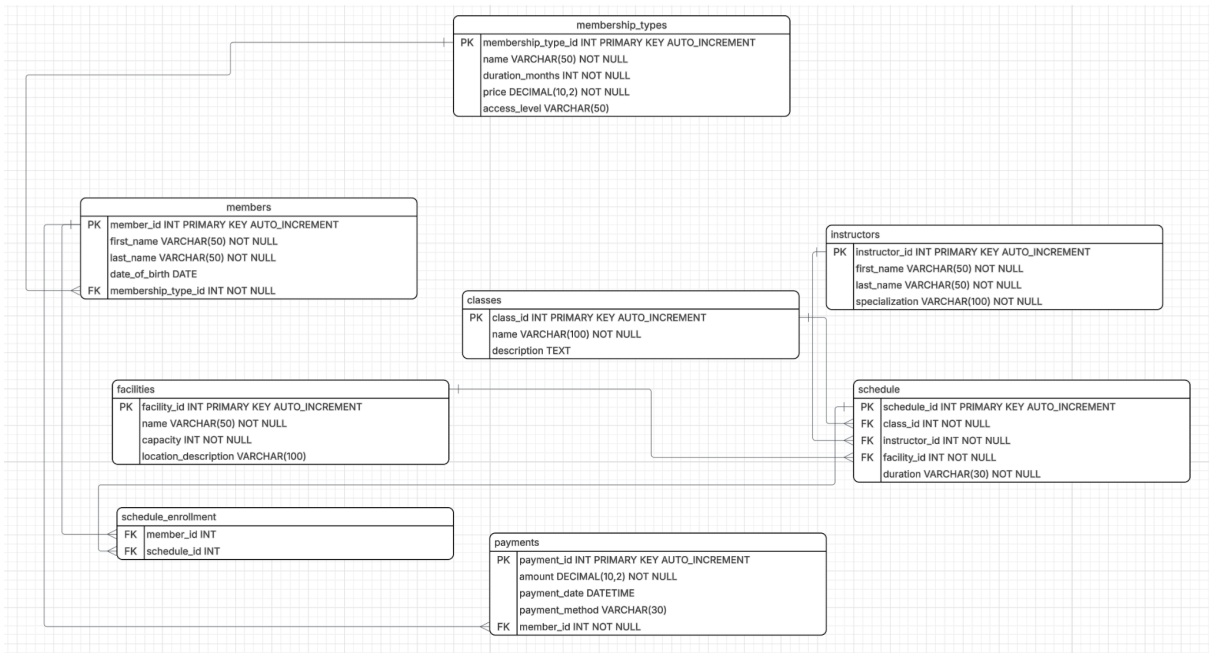


1.1 Logical Scheme



1.2 Objects

Table Description (Members)

The central repository for all registered clients, storing their personal details and linking them to a specific membership plan.

Table Name	Field name	Field Description	Data Type / Constraints
Members	member_id	Unique identifier for each gym member.	Int, PK, Auto increment
	first_name	The member's given name.	VAR(50),not null
	last_name	The member's family name.	VAR(50),not null

	date_of_birth	The member's birth date.	DATE
	membership_type_id	Reference to the specific plan the member has purchased.	INT, Not null, FK

- Comments on table relationships: Many-to-One with **membership_types** (each member follows one plan).
 - One-to-Many with **payments** (one member can have multiple transaction records).
 - One-to-Many with **schedule_enrollment** (one member can sign up for multiple scheduled sessions).

member_id	first_name	last_name	date_of_birth	membership_type_id
101	John	Doe	1990-05-15	1

Table Description (Membership_types)

A lookup table (dictionary) that stores the available subscription plans offered by the gym, including their costs, durations, and specific access permissions.

Table Name	Field name	Field Description	Data Type / Constraints
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membership_types	membership_type_id	Unique identifier for each membership plan.	Int, PK, Auto increment
	name	The name of the membership tier (e.g., "Basic", "Premium").	VAR(50),not null
	duration_months	The validity period of the membership in months.	int,not null
	price	The cost of the membership plan.	Decimal(10,2),not null
	access_level	Defines which areas or services the member can use.	var(50)

- Comments on table relationships: This is a parent table to **members**. It has a One-to-Many relationship with the **members** table, as one membership type can be assigned to many different members.

membership_type_id	name	duration_months	price	access_level
1	Gold Membership	12	599.99	Full Access

Table Description (Classes)

A catalog of the workout programs and services offered (e.g., Yoga, HIIT), defining what each class entails without specific timing.

Table Name	Field name	Field Description	Data Type / Constraints
classes	class_id	Unique identifier for a type of workout program.	Int, PK, Auto increment
	name	The title of the class (e.g., "Morning Yoga").	VAR(100),not null
	description	A summary of what the class involves.	text

Comments on table relationships: Has a One-to-Many relationship with the [schedule](#) table. One class definition (e.g., "Zumba") can appear multiple times in the weekly schedule.

class_id	name	description
20	Vinyasa Flow	A dynamic yoga class focusing on breath and movement.

Table Description (Facilities)

An inventory of the gym's physical spaces (studios, pools, gyms) used to manage maximum occupancy and room locations.

Table Name	Field name	Field Description	Data Type / Constraints
facilities	facility_id	Unique identifier for a specific room or area.	Int, PK, Auto increment
	name	The name of the room (e.g., "Studio A", "Pool").	VAR(50),not null
	capacity	Maximum number of people allowed in the area at once.	Int,not null
	location_description	Details on where the facility is located within the building.	VAR(100)

Comments on table relationships: Has a One-to-Many relationship with the [schedule](#) table. A single facility can host many different scheduled classes throughout the day.

facility_id	name	capacity	location_description
5	Studio A	25	Second floor, near the elevators

Table Description (Payments)

A financial log that records every transaction made by members, used to track revenue and individual payment history.

Table Name	Field name	Field Description	Data Type / Constraints
payments	payment_id	Unique identifier for each transaction.	Int, PK, Auto increment
	amount	Total money paid in the transaction.	DECIMAL(10,2),NOT NULL
	payment_date	Timestamp of when the payment was processed.	DATETIME
	payment_method	The mode of payment (e.g., "Credit Card", "Cash").	VAR(30)
	member_id	Reference to the member who made the payment.	INT, not null,FK

- Comments on table relationships: Has a Many-to-One relationship with **members**. Every payment must be linked to a specific member ID to track who paid the amount.

payment_id	amount	payment_date	payment_method	member_id
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5001	599.99	2024-01-10 14:30:00	Credit Card	101
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Table Description (Schedule)

The operational heart of the system that schedules a specific class in a specific room with a specific instructor at a set time.

Table Name	Field name	Field Description	Data Type / Constraints
schedule	schedule_id	Unique identifier for a specific time slot in the calendar.	Int, PK, Auto increment
	class_id	Reference to the program being taught.	INT,not null,FK
	instructor_id	Reference to the trainer leading this session.	INT,not null,FK
	facility_id	Reference to the room where the session takes place.	INT,not null,FK
	duration	The specific time or length of the session.	VAR(30), not null

Comments on table relationships: * Many-to-One with [classes](#), [instructors](#), and [facilities](#).

- One-to-Many with **schedule_enrollment** (one scheduled slot can have many members signed up for it).

schedule_id	class_id	instructor_id	facility_id	duration
301	20	10	5	Mon 10:00 - 11:00

Table Description (Instructors)

A personnel directory containing names and professional specializations of the trainers employed at the facility.

Table Name	Field name	Field Description	Data Type / Constraints
instructors	instructor_id	Unique identifier for each staff trainer.	Int, PK, Auto increment
	first_name	The instructor's given name.	VAR(50),not null
	last_name	The instructor's family name.	VAR(50),not null
	specialization	The trainer's area of expertise (e.g., "Yoga", "HIIT").	VAR(30),not null

- Comments on table relationships: Has a One-to-Many relationship with the **schedule** table. One instructor can be assigned to lead many different sessions in the schedule.

instructor_id	first_name	last_name	specialization
10	Sarah	Jenkins	Yoga

Table Description (Schedule_enrollment)

A junction table that manages the Many-to-Many relationship between members and the schedule, acting as the official registration list for every session.

Table Name	Field name	Field Description	Data Type / Constraints
schedule_enrollment	member_id	Reference to the member who is signing up.	INT,FK,PRIMARY KEY (member_id, schedule_id)
	schedule_id	Reference to the specific session time chosen by the member.	INT,FK,PRIMARY KEY (member_id, schedule_id)

- Comments on table relationships: This table facilitates the Many-to-Many link between **members** and **schedule**. It uses a Composite Primary Key consisting of **member_id** and **schedule_id**. Both fields act as Foreign Keys pointing back to their respective parent tables.

member_id	schedule_id
101	301

